

Dear Editors,

Please find enclosed our manuscript entitled “Interband Berry connection measurement in the optical honeycomb lattice” which we are submitting for consideration for publication in *Nature*.

Placing this work in the context of high impact publications in *Nature* and other Springer journals

In this manuscript, we show for the first time that the interband Berry connection, a measure of the relative geometry of two Bloch bands, can be measured directly in experiments through its fundamental connection to the optical response of materials. Many of the most impactful recent discoveries in condensed-matter physics have revealed how physical properties of materials derive from geometric properties of Bloch bands. Understanding the importance of Bloch band geometry has stirred interest in experimental methods that could measure these geometric properties. Several prominent works have shown how, in certain specialized systems and relying on foreknowledge of band structure, the Bloch states across the Brillouin zone can be characterized experimentally through a form of state tomography Asteria et al. (2019); Gianfrate et al. (2020); Kang et al. (2025); Guillot et al. (2025). From such tomographic reconstruction, one can calculate the differential variation of Bloch states with quasimomentum, obtaining geometric measures such as the interband Berry connection and the complete quantum geometric tensor.

In our work, we take a different track that provides a more natural and general approach to measuring band geometry. Specifically, we make use of the direct connection between the response of materials to optical excitation and the interband Berry connection, which characterizes the relative geometry of two bands that are connected by an optical transition. By resolving the optical response in quasimomentum space, the interband Berry connection can be measured directly with no further assumptions about the Bloch-state wavefunction. We demonstrate this approach using an ultracold-atom quantum simulator, one in which electrons in solids are replaced with quantum degenerate Fermi gases in an optical lattice, and in which optical excitation is replaced by lattice-shaking spectroscopy. Taking the two-dimensional honeycomb lattice as a test case, we experimentally visualize detailed features of the interband Berry connection. Most notably, we observe the necessary presence of Dirac strings in the interband Berry connection between two bands of the honeycomb lattice (bands 1 and 3).

We feel our manuscript uniquely brings the relation between linear optical response and the geometry of band structure to light. The method we demonstrate to measure the interband Berry connection through quasimomentum-resolved response to optical excitation at various polarizations is broadly applicable, not just across the wide variety of optical lattice systems being explored worldwide, but also potentially in solids.

We note, further, that much of the past exploration of band geometry and its implication to material properties has focused on “intraband” effects, where the geometry and topology of just a single band is at play. A famous example of such single-band phenomenology is the classification of topological insulators. More recently, attention is being turned to multi-band geometric features, such as the Euler class that typifies two neighboring bands in lattices of certain symmetry. Our approach to measuring the relative geometry of two bands, through the interband (or off-diagonal) Berry connection, will be an important experimental tool for such multi-band geometry and topology explorations.

Achievements of our work

To our knowledge, this work is the first direct extraction of the interband Berry connection, an element of the quantum geometric tensor, using momentum resolved optical excitations. Building upon the publications from Springer journals referenced above, this work combines the momentum space resolution of ARPES with the measurement of the optical excitation response between different bands (as opposed to a measurement of particles expelled from a sample like in ARPES). These measurements are significant as they connect the elusive quantum geometric object, the interband Berry connection, to the response of a crystal to optical excitations in a clean, controllable system. In addition, our measurements reveal

features that haven't been observed before, like a winding of the interband Berry connection around the Brillouin zone center in the absence of any corresponding band touching points, or their investigation has been skipped entirely, like the role of the Dirac string in the interband Berry connection.

While our system does not feature a special topological setting, this keeps our findings, for example quasimomentum dependent selection rules, applicable to a broad class of systems.

What's coming after our work?

We believe our work will have a significant impact on other fields. We complement the measurement capabilities of solid-state physicists by having the ability to observe the response to optical interband excitations with high momentum resolution. Though experimentally challenging, our work incorporates all ingredients for the study of the full Euler connection in optical lattices with the right unit-cell structure and symmetry (such as the kagome lattice). Another natural extension of our work will be the quantum simulation of not just linear but also nonlinear optics in crystals. For example, the nonlinear optical susceptibility in solids should map onto nonlinear mechanical susceptibility in ultracold-atom systems. Investigating the nonlinear optical response of crystalline materials with full quasimomentum resolution would open the door to many novel studies.

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. We confirm that there are no known conflicts of interest associated with publication of this article, and there has been no financial support for this work that has swayed its outcome. All authors have approved the manuscript and agree with its submission to *Nature*. We would like to suggest the following reviewers for our manuscript: Nur Ünal, Monika Aidelsburger, and Bohm-Jung Yang. Thank you for considering our manuscript for publication. We look forward to hearing from you soon.

References

- Luca Asteria, Duc Thanh Tran, Tomoki Ozawa, Matthias Tarnowski, Benno S. Rem, Nick Fläschner, Klaus Sengstock, Nathan Goldman, and Christof Weitenberg. Measuring quantized circular dichroism in ultracold topological matter. *Nature Physics*, 15(5):449–454, May 2019. ISSN 1745-2481. doi: 10.1038/s41567-019-0417-8. URL <https://www.nature.com/articles/s41567-019-0417-8>.
- A. Gianfrate, O. Bleu, L. Dominici, V. Ardizzone, M. De Giorgi, D. Ballarini, G. Lerario, K. W. West, L. N. Pfeiffer, D. D. Solnyshkov, D. Sanvitto, and G. Malpuech. Measurement of the quantum geometric tensor and of the anomalous Hall drift. *Nature*, 578(7795):381–385, February 2020. ISSN 1476-4687. doi: 10.1038/s41586-020-1989-2. URL <https://www.nature.com/articles/s41586-020-1989-2>.
- Martin Guillot, Cédric Blanchard, Martina Morassi, Aristide Lemaître, Luc Le Gratiet, Abdelmounaim Harouri, Isabelle Sagnes, Robert-Jan Slager, F. Nur Ünal, Jacqueline Bloch, and Sylvain Ravets. Measuring non-Abelian quantum geometry and topology in a multi-gap photonic lattice, November 2025. URL <http://arxiv.org/abs/2511.03894>. arXiv:2511.03894 [cond-mat].
- Mingu Kang, Sunje Kim, Yuting Qian, Paul M. Neves, Linda Ye, Junseo Jung, Denny Puntel, Federico Mazzola, Shiang Fang, Chris Jozwiak, Aaron Bostwick, Eli Rotenberg, Jun Fuji, Ivana Vobornik, Jae-Hoon Park, Joseph G. Checkelsky, Bohm-Jung Yang, and Riccardo Comin. Measurements of the quantum geometric tensor in solids. *Nature Physics*, 21(1):110–117, January 2025. ISSN 1745-2481. doi: 10.1038/s41567-024-02678-8. URL <https://www.nature.com/articles/s41567-024-02678-8>.